

Assignment

Explain all conditional statements with examples

Ans: Conditional Statements in Python

Conditional statements are used to make decisions in a program. They execute different blocks of code based on whether a condition is **True** or **False**.

1. if Statement

The `if` statement executes a block of code only if the condition is true.

Syntax

```
if condition:
    statement
```

Example

```
age = 18

if age >= 18:
    print("You are eligible to vote.")
```

Output:

You are eligible to vote.

2. if-else Statement

The `if-else` statement executes one block if the condition is true and another block if it is false.

Syntax

```
if condition:
    statement1
else:
    statement2
```

Example

```
num = 5

if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

Output:

Odd Number

3. if-elif-else Statement

The `elif` (else if) statement checks multiple conditions one by one.

Syntax

```
if condition1:
    statement1
elif condition2:
    statement2
else:
    statement3
```

Example

```
marks = 75

if marks >= 90:
    print("Grade A")
elif marks >= 70:
    print("Grade B")
elif marks >= 50:
```

```
        print("Grade C")
else:
    print("Fail")
```

Output:

Grade B

4. Nested if Statement

An if statement inside another if statement is called a nested if.

Example

```
age = 20
has_license = True

if age >= 18:
    if has_license:
        print("You can drive.")
    else:
        print("Get a driving license first.")
```

Output:

You can drive.

5. Short-Hand if Statement

Used when there is only one statement to execute.

Example

```
x = 10

if x > 5: print("x is greater than 5")
```

Output:

x is greater than 5

6. Short-Hand if-else (Ternary Operator)

Example

```
a = 10
b = 20

print("A is greater") if a > b else print("B is greater")
```

Output:

B is greater